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Bihar Engineering University, Patna

B. Tech. I st Semester Examination, 2024

Time: 03 Hours

Code: 100102 ***Subject: Engineering Mathematics -I*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option the following (Any seven question only): **[14]**

(a) Which operation is not an elementary row operation?
- (i) Swapping two rows
- (ii) Adding a multiple of one row to another
- (iii) Multiplying a row by zero.
- (iv) Multiplying a row by a non-zero constant

(b) The rank of a matrix is:
- (i) Number of non-zero rows in echelon form
- (ii) Number of columns
- (iii) Number of rows
- (iv) None above

(c) The inverse of a matrix using Gauss-Jordan:
- (i) only determinant calculation
- (ii) Reducing matrix to diagonal form
- (iii) Augmenting with identity matrix and applying row operations
- (iv) None of the above

(d) A matrix is diagonalizable:
- (i) It has repeated eigenvalues only
- (ii) It is symmetric
- (iii) It has a full set of linearly independent eigenvectors
- (iv) It is square

(e) A matrix is similar to a diagonal matrix if:
- (i) It has complex entries
- (ii) It has n linearly independent eigenvectors
- (iii) It is invertible
- (iv) None of the above

(f) A function is Riemann integrable if:
- (i) It is continuous
- (ii) It has infinite discontinuities
- (iii) It is discontinuous everywhere
- (iv) None of the above

(g) Jacobian is:
- (i) A scalar
- (ii) A vector
- (iii) A matrix of partial derivatives
- (iv) A function

(h) Taylor's expansion for multivariable function includes:
- (i) Only first order terms
- (ii) Only second order
- (iii) All orders
- (iv) Only constant term

(i) Change of variables in double integral involves:
- (i) Matrix algebra
- (ii) Laplace transforms
- (iii) Jacobian
- (iv) Only constant term

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(j) Triple integral over a region gives:

(i) Area (iii) Volume

(ii) Length (iv) Angle

Q.2 Solve both questions : **[14]**

(a) What is the difference between a vector space and a subspace? Illustrate with examples.

(b) Define and explain rank, row space, and column space of a matrix with examples.

Q.3 Solve both questions : **[14]**

(a) Explain the Cayley-Hamilton Theorem and verify it for a 2x2 matrix.

(b) Define and compute the Jacobian for transformation $x = u + v$ of $u - v$.

Q.4 Solve both questions : **[14]**

(a) Explain the Gauss-Jordan method for finding the inverse of a matrix with an example.

(b) Define Hermitian, Skew-Hermitian, and Unitary matrices. Provide examples for each.

Q.5 Solve both questions : **[14]**

(a) Explain and prove the Rank-Nullity theorem.

(b) Describe the method of finding eigenvalues and eigenvectors of a matrix.

Q.6 Solve both questions : **[14]**

(a) State and prove Rolle's Theorem with a graphical example.

(b) Define Beta and Gamma functions. Derive the relation between them.

Q.7 Solve both questions : **[14]**

(a) Derive the Taylor series for a function of two variables.

(b) Solve a maxima-ratum problem using second derivative test.

Q.8 Solve both questions : **[14]**

(a) Evaluate a double integral to find area under a curve.

(b) Convert a double integral into polar coordinates and evaluate.

Q.9 Write short notes on any two of the following: **[14]**

(a) Properties of Eigen vectors

(b) Riemann Integration & Riemann Sum

(c) Application to area and volume using double and triple integral

(d) Scalar and vector fields

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